Literature Highlights

Near-Infrared Camera

Silent Speech Interface System

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1. Understanding near infrared spectroscopy and its application to skeletal muscle research[1]

This paper provides an examination of near infrared spectroscopy (NIRS) and its utilization in skeletal muscle research. NIRS functions by measuring the relative concentrations of hemoglobin (Hb) and myoglobin (Mb) in tissues, which are pivotal indicators of muscle oxygenation and metabolic activity.

NIRS principles described mathematical equations like Beer-Lambert Law and modified version. This equation outline the balance of oxygen consumption and delivery in muscle tissues. Central to this equation are the aspects of light absorption and scattering, factors influenced by skin melanin and adipose tissue thickness (ATT). The paper details how different chromophores, particularly Hb and Mb, absorb light at specific wavelengths (750-840nm), allowing NIRS to quantify changes in muscle oxygenation.

$$\mathbf{A}\ (\mathrm{OD}) = \mathbf{Log}\left(\frac{I_0}{I}\right) = \varepsilon[c]\mathbf{L}\cdot\mathbf{DPF} + \mathbf{G}$$

Figure 1. Modified Beer-Lambert Law

The paper also highlights the implementation of novel NIRS technologies, including continuous wave, time domain and frequency domain spectrometers, which offer improved resolution and deeper tissue penetration. These advancements are proposed to expand the applicability of NIRS across different research domains, offering insights into the dynamic nature of muscle physiology. The review also addresses various sources of variability in NIRS measurements, such as tissue thickness and instrument calibration, to enhance the accuracy.

Relevance:

1. NIRS monitoring muscle activity informs the utilization of NIR technology for detecting laryngeal muscle activity in silent speech interface (SSI).
2. NIRS detect muscle activity through [Hb] and [Mb] supports the transition from EMG methods to NIR cameras.

2. Correspondence between laryngeal vocal fold movement and muscle activity during speech and nonspeech gestures[2]

The paper provides the correlation between laryngeal vocal fold movement and muscle activity during various speech and non-speech gestures. EMG is utilized to record

activation of four laryngeal muscles: the posterior cricoarytenoid (PCA), lateral cricoarytenoid (LCA), cricothyroid (CT), and thyroarytenoid (TA) in four subjects during tasks such as sniffing, coughing, and speech. Using video-assisted nasolaryngoscopy for accurate visualization, EMG signals were synchronized with vocal fold movements, enabling precise measurements of the onset and offset of muscle activity in relation to vocal fold actions.

Paper compute correlation coefficients for each muscle and task to evaluate muscle activation patterns and their effects on vocal fold. Notably, muscle activity showed correlations during tasks involving vocal fold opening and closing. PCA opens the vocal folds rapidly by rotating the arytenoid cartilages outward to swallow and produce voiceless sound. LCA brings the vocal folds together for phonation. CT lengthen the vocal folds and TA shorten the vocal folds, which changing pitch of voice. The delay between muscle contraction onset and the resultant vocal fold movement are important in laryngeal dynamics.

Relevance:

1. Examination of muscle behavior and timing relative to vocal fold movement.
2. The design and calibration of sensing systems could capture laryngeal muscle dynamics for accurate silent speech analysis.
3. Reduce interruption of large larynx movement when collecting data and testing.

3. TieLent: A Casual Neck-Mounted Mouth Capturing Device for Silent Speech Interaction[3]

This paper introduces a SSI named TieLent utilizes a neck-mounted RGB camera, strategically positioned below the user's chin with different angle to capture lip movements.

The device was tested using 34 words and 127 phoneme with accuracy of approximately 94%. This approach leverages image-to-speech neural networks, converting visual mouth movements into audible speech by convolutional neural network (CNN) layers to transform 32x32 grayscale lip images into Mel-frequency cepstral coefficients (MFCCs), which are standard in audio feature vector extraction.

To solve the challenge of limited data for deep learning, an autoencoder was employed to perform unsupervised learning on a small dataset. The autoencoder compressed input features, allowing it to learn essential feature representations in reduced dimensions. Using the LRS2 dataset, which offers varied facial appearances, the convolutional layers of the autoencoder were trained to abstract lip features effectively. These pre-trained

layers were transferred to the image-to-speech network, facilitating improved learning efficiency. Subsequent training was conducted without freezing weights, enabling dynamic adaptation.

Relevance:

1. Provides an imaging techniques with neural networks to convert visual data into phonetic representations.
2. Leverage similar neural network architectures and image processing methods can interpret muscle activity as speech commands.

4. Speaking without vocal folds using a machine-learning-assisted wearable sensing-actuation system[4]

This paper provide a wearable magnetoelasticity sensing SSI. It capture extrinsic laryngeal (Sternothyroid) muscle movement and convert them into 3-dimensional electrical signals, which can be translated into speech signals.

The device is constructed using poly-dimethylsiloxane (PDMS), embedded with serpentine copper coils. These coils operate within a magnetically active matrix to detect muscle deformation in three dimensions. The system utilize the piezoelectric properties and kirigami structure producing a robust signal-to-noise ratio. The device has been proven to detect distinguishable signals of each syllabus from the laryngeal muscle movement without losing any essential waveform characteristics for downstream analysis.

The machine learning utilize a two-step algorithm involving training and classification. The process starts with filtering the electrical signals from muscle movements, which are compacted using nth-order matrices for principal component analysis (PCA) to reduce redundancy. Multi-class support vector classification (SVC) is applied to categorize the signals, employing a "one vs. rest" decision function to increase sentence accuracy. After training with data, real-time signals are classified to sentence.

Relevance:

1. Indicate a feasible detection of laryngeal muscle only through the utilization of classify machine learning architectures.
2. A clear pipeline of classification ML is provided

References

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